

DIAGONAL: Entity-Grounded Topological Evaluation for Multi-Shot Video Narrative Compliance

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Abstract

Text-to-multi-shot video (T2MSV) generation models are evaluated primarily on visual quality and frame-level consistency, yet these metrics fail to capture whether models can execute *narrative dynamics*—the structured entry and exit of entities across shots. We formalize multi-shot narratives as *state transition matrices* $\Delta S \in \{-1, 0, +1\}^{(T-1) \times N}$, where each element encodes an atomic operation: entry (+1), exit (−1), or hold (0). We prove that all possible 3-shot narrative topologies can be spanned by just two basis matrices under column permutation and linear combination (**Theorem 1**), establishing mathematical completeness. Building on this, we introduce DIAGONAL, a benchmark comprising 1,000 prompts spanning 7 narrative patterns (3 basis, 4 derived). A vision-language model constructs the observed presence matrix S_{obs} via per-entity binary queries, yielding a clean matrix-matching pipeline: the *Prescribed Transition Match* (PTM) measures the fraction of structurally prescribed transitions ($\Delta S \neq 0$ cells) correctly observed in the generated video. Evaluation of four SOTA T2MSV architectures across 3,600 videos reveals statistically significant method rankings (Kruskal–Wallis $H = 275$, $p < 10^{-59}$): StoryDiffusion achieves the highest PTM (0.243) while VIC scores lowest (0.070), with all pairwise comparisons significant ($p < 0.005$) and non-overlapping bootstrap confidence intervals. We identify a *coherence–compliance Pareto dilemma* ($r_{\text{IC-PTM}} = -0.998$) and four systematic failure types, providing actionable architectural diagnostics.

CCS Concepts

• Computing methodologies → Computer vision; Neural networks.

Keywords

Multi-Shot Video Generation, Narrative Evaluation, State Transition Matrix, Video Benchmark

ACM Reference Format:

Anonymous Author(s). 2026. DIAGONAL: Entity-Grounded Topological Evaluation for Multi-Shot Video Narrative Compliance. In . ACM, New York, NY, USA, 9 pages. <https://doi.org/10.1145/nnnnnnnn.nnnnnnnn>

1 Introduction

Diffusion-based generative models [14, 35] have rapidly expanded from single-clip to text-to-multi-shot video (T2MSV) generation [3, 13, 16]. Recent systems such as StoryDiffusion [58], EchoShot [45], VIC [7], and VGoT [46] generate temporally connected sequences of scenes, leveraging consistent self-attention, flow-matching, in-context learning, and cascaded T2I-to-I2V pipelines. These systems have demonstrated impressive visual fidelity in recent evaluations [18,

19], and commercial systems like Sora [4] and Movie Gen [33] suggest rapid progress toward production-quality video generation.

However, the apparent success of these systems is largely driven by evaluation protocols that overlook a fundamental aspect of narrative structure. Existing benchmarks [18, 19, 25, 38] evaluate identity consistency, visual quality, or single-clip compositionality, but none systematically test whether models can execute *narrative dynamics*: the structured introduction, co-occurrence, and removal of entities across shots. This gap is critical because real-world video narratives—from films [25] to advertisements to educational content—are fundamentally defined by *who appears, when, and alongside whom*. A scene where two characters meet and then part ways conveys entirely different meaning from one where they remain together throughout; yet both would receive identical consistency scores. Without metrics that capture these dynamics, we cannot meaningfully assess whether T2MSV models are progressing toward their intended purpose, and the community risks optimizing toward the wrong objective.

Consider a relay narrative: $A \rightarrow AB \rightarrow B$, where entity A appears alone, both co-occur, then only B remains. This seemingly simple pattern requires three distinct capabilities: (1) introducing B while preserving A ’s appearance and identity, (2) maintaining co-occurrence with both entities visually distinguishable, and (3) removing A while B persists in isolation. Current models systematically fail at (3)—the departed entity “bleeds” into subsequent shots (**Context Bleeding**), a failure mode invisible to existing metrics. Critically, Identity Consistency (IC) [20] *rewards* this failure because persistent entities increase cross-shot similarity. We term this the **consistency paradox**: IC is adversarial to narrative compliance for any non-trivial pattern. This paradox has far-reaching implications: the community may be inadvertently optimizing models *away from* narrative competence by chasing IC improvements, creating a systematic blind spot in the evaluation landscape.

To address these limitations, we formalize entity dynamics as a *state transition matrix* (a compact encoding of which entities enter, exit, or persist at each shot boundary) and prove that **all** 3-shot topologies are generated from just two basis matrices (**Theorem 1**). Our evaluation reveals a *coherence–compliance Pareto dilemma*: no current architecture simultaneously achieves high visual coherence and high narrative compliance, suggesting a fundamental architectural tension that the community must resolve. DIAGONAL contributes:

- A **topological formalization** with mathematical completeness guarantee for multi-shot narrative structures (§4).
- A **VLM-based matrix-matching pipeline** that constructs observed presence matrices via binary entity queries and evaluates them against prescribed transition matrices (§5).
- A **1,000-prompt benchmark** spanning 7 patterns with a primary metric (PTM) achieving statistically significant discrimination across all method pairs ($p < 0.005$) (§7).

Table 1: Benchmark comparison. $\boxtimes$ supported, $\times$ not, $\triangle$ partial.

Benchmark	Multi-shot	Entity Dec.	Absence	Formal.	Fail. Tax.
VBench 2.0 [19]	$\times$	$\times$	$\times$	$\times$	$\times$
T2V-CompBench [38]	$\times$	$\triangle$	$\times$	$\times$	$\times$
MovieBench [25]	$\boxtimes$	$\triangle$	$\times$	$\times$	$\times$
LoCoT2V [56]	$\boxtimes$	$\times$	$\times$	$\times$	$\times$
ShotAdapter [20]	$\boxtimes$	$\triangle$	$\times$	$\times$	$\times$
DIAGONAL (Ours)	$\boxtimes$	$\boxtimes$	$\boxtimes$	$\boxtimes$	$\boxtimes$

- A **four-type failure taxonomy** providing actionable architectural diagnostics (§5.5).
- The **consistency paradox**: holistic metrics are adversarial to narrative evaluation ($\eta_{\text{IC-PTM}} = -0.998$) (§3).

2 Related Work

Text-to-video generation. The foundation of modern video synthesis rests on diffusion models [14, 35, 36], extended temporally through UNet attention [3, 13, 16], scalable DiT architectures [4, 31, 44] with transformer backbones [6, 42], flow-matching [26], and classifier-free guidance [15]. Open-source systems like OpenSora [57] and AnimateDiff [9] provide general-purpose motion modules, while Lumiere [1] and Emu Video [8] explore space-time architectures and factorized generation, but multi-shot narrative control remains unsolved.

Multi-shot video generation. Multi-shot extensions address coherence via shared self-attention [22, 58], cross-attention [28], latent initialization [29], LLM-guided layouts [23, 46, 48], keyframe transitions [5, 52], flow-matching portraits [45], IP-Adapter [55], in-context LoRA [7, 17], subject customization [49, 51], and multimodal storytelling [37, 53]. Longer-horizon generation has been explored via autoregressive token models [21, 43] and visual pre-training [50], though these operate at the single-clip rather than multi-shot level. These span four paradigms: CSA (StoryDiffusion [58]), flow-matching (EchoShot [45]), in-context DiT (VIC [7]), and cascaded T2I+I2V (VGoT [46]). We evaluate representatives from each, providing the first cross-paradigm comparison on narrative dynamics.

Video generation benchmarks. Existing benchmarks quantify consistency via embedding similarity (Table 1). VBench [18, 19] computes frame-level CLIP [34]/DINO [30] similarities conflating subject and background; T2V-CompBench [38] targets single-clip compositionality. ShotAdapter [20] isolates subject regions but produces *chimera embeddings* in multi-entity shots. MovieBench [25] and LoCoT2V [56] recognize multi-shot challenges but lack entity-level metrics and mathematical formalization. FID [12] and FVD [41] measure distributional fidelity but are agnostic to narrative structure; CLIPScore [11] measures text-image alignment but only at the single-frame level. DIAGONAL is the first to formalize multi-shot narratives via state transition matrices with completeness guarantees.

3 The Consistency Paradox

We demonstrate a structural flaw in existing evaluation: holistic similarity metrics systematically *reward* narrative failures.

ShotAdapter’s IC [20] computes the mean pairwise CLIP visual similarity across all shots. For any narrative pattern where at least one entity enters or exits (i.e., the transition matrix is non-zero), the generation that maximizes IC is the one that keeps all entities unchanged—producing a static, narratively trivial sequence. For the relay pattern ($A \rightarrow AB \rightarrow B$), a model that incorrectly preserves A in all shots achieves $\text{IC} \approx 0.94$, while one that correctly removes A from Shot 3 achieves $\text{IC} \approx 0.78$ —a 17% penalty for correct behavior.

We validate this across 3,600 videos generated by four models: IC and our PTM metric (§5.3) produce opposite rankings at the model level ($\eta_{\text{IC-PTM}} = -0.998$). VIC achieves highest IC (0.94) but lowest PTM (0.070); StoryDiffusion has highest PTM (0.243) but lower IC (0.81). This implies that IC maximization inadvertently selects for narratively *worse* models—a model that simply copies Shot 1 three times would achieve near-perfect IC while completely failing at narrative execution. The paradox motivates the matrix-matching evaluation we introduce next.

4 Topological Formalization

In this section, we formalize the notion of “narrative dynamics” and prove that a small set of basis patterns is sufficient to represent *all possible* multi-shot narrative structures.

4.1 Notation and Definitions

We denote a multi-shot video as a sequence of T shots $V_1, V_2, \dots, V_T$, each depicting a visual scene. Each video involves N named entities $e_1, e_2, \dots, e_N$ (e.g., “a chef,” “a dog”). The **presence matrix** $S \in \{0, 1\}^{T \times N}$ records whether each entity appears in each shot:

$$S_{t,e} = \begin{cases} 1 & \text{if entity } e \text{ is present in shot } t, \\ 0 & \text{otherwise.} \end{cases}$$

For example, the relay narrative $A \rightarrow AB \rightarrow B$ has:

$$S = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad (\text{rows} = \text{shots, columns} = \text{entities}).$$

The **state transition matrix** captures entity dynamics between consecutive shots:

$$\Delta S = S_{2:T} - S_{1:T-1}, \quad (1)$$

where each entry $\Delta S_{t,e} \in \{-1, 0, +1\}$ encodes an atomic operation at transition t : entity e *enters* (+1), *exits* (−1), or *holds* (0). For the relay example: $\Delta S = \begin{bmatrix} 0 & +1 \\ -1 & 0 \end{bmatrix}$, meaning B enters at transition 1 and A exits at transition 2.

We focus on $T = 3$ shots with $N \leq 3$ entities, yielding $\Delta S \in \{-1, 0, +1\}^{2 \times 3}$. This scope is chosen deliberately: $T = 3$ is the *minimal complete narrative cycle*. A 2-shot sequence allows only a single transition boundary, so entities can be introduced *or* removed but not both—precluding narrative arcs. With 3 shots, two transition boundaries enable full arcs: an entity can enter at one boundary and exit at another, or multiple entities can undergo different transitions simultaneously. The constraint $N \leq 3$ is also practical:

the vast majority of multi-shot video prompts involve 2–3 named entities, and the algebraic structure generalizes naturally to $N > 3$ via additional columns.

4.2 Basis Patterns

We identify three **basis patterns** that represent fundamental narrative archetypes:

Relay ($\mathcal{B}_1$): $A \rightarrow AB \rightarrow B$. One entity hands off to another— B enters while A exits: $\mathcal{B}_1 = \begin{bmatrix} 0 & +1 \\ -1 & 0 \end{bmatrix}$.

Split ($\mathcal{B}_2$): $AB \rightarrow A \rightarrow B$. Co-occurring entities separate into alternating scenes: $\mathcal{B}_2 = \begin{bmatrix} 0 & -1 \\ -1 & +1 \end{bmatrix}$.

Accumulation ($\mathcal{B}_3$): $A \rightarrow AB \rightarrow ABC$. Entities are progressively introduced: $\mathcal{B}_3 = \begin{bmatrix} 0 & +1 & 0 \\ 0 & 0 & +1 \end{bmatrix}$.

Four **derived patterns** are obtained by applying time-reversal (negating and reversing rows of ΔS) and column permutation (swapping entity roles):

Reduction ($ABC \rightarrow AB \rightarrow A$): entities are progressively removed, the time-reversal of Accumulation. $\Delta S = \begin{bmatrix} 0 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix}$.

Convergence ($A \rightarrow B \rightarrow AB$): two separately introduced entities eventually co-occur. $\Delta S = \begin{bmatrix} -1 & +1 \\ +1 & 0 \end{bmatrix}$.

Reverse Relay ($B \rightarrow AB \rightarrow A$): time-reversal of Relay— A enters as B exits. $\Delta S = \begin{bmatrix} +1 & 0 \\ 0 & -1 \end{bmatrix}$.

Sliding Window ($AB \rightarrow BC \rightarrow C$): the most demanding pattern with 4 non-zero transitions—simultaneous exit of A and entry of C , followed by exit of B . $\Delta S = \begin{bmatrix} -1 & 0 & +1 \\ 0 & -1 & 0 \end{bmatrix}$.

Table 2 lists all seven patterns with their properties.

4.3 Completeness Theorem

The central theoretical contribution of this paper is showing that the basis patterns are not merely representative—they *span the entire narrative space*.

THEOREM 4.1 (COMPLETENESS OF THE NARRATIVE SPACE). *Under column permutation and integer-coefficient linear combination, the two basis patterns $\mathcal{B}_1$ (Relay) and $\mathcal{B}_2$ (Split) generate all matrices in $M_{2 \times 3}(\mathbb{Z})$.*

Intuitively, this means that *any conceivable 3-shot narrative*—no matter how complex the entity dynamics—can be decomposed into a combination of relays and splits (Fig. 1).

PROOF SKETCH. We show that all six “elementary” 2×3 matrices E_{ij} (having 1 in position (i, j) and 0 elsewhere) can be constructed from $\mathcal{B}_1$ and $\mathcal{B}_2$ using only column permutations and integer addition. Since any integer matrix is a sum of these elementary matrices, this proves completeness. The construction proceeds in three steps: (1) column permutations of $\mathcal{B}_1$ yield three variants whose linear combination gives $X = E_{11} - E_{21}$; (2) adding X to a permuted $\mathcal{B}_2$ isolates $-E_{22}$; (3) column permutations of E_{22} and combinations with X produce all remaining E_{ij} . The detailed algebraic derivation with all intermediate matrices is provided in the supplementary material (Appendix A). $\square$

Implications. Theorem 1 has three important consequences. *First*, DIAGONAL’s seven patterns form a *mathematically complete* basis: evaluating a model on these patterns is sufficient to characterize its behavior on *any* narrative topology, including patterns not explicitly in the benchmark. This distinguishes our framework

How two bases span all narratives:

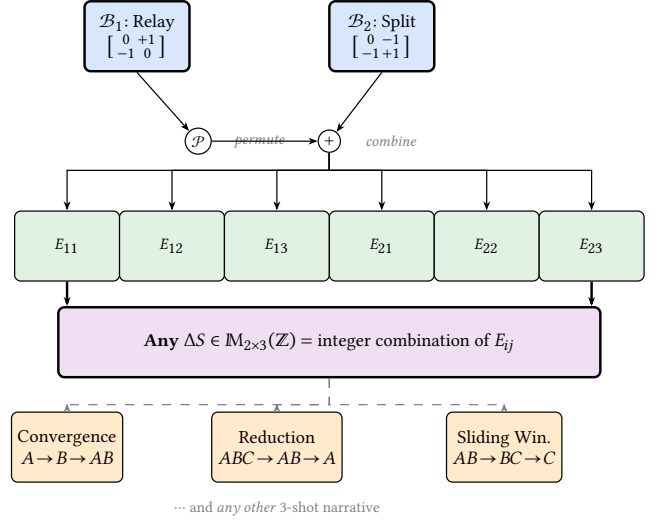


Figure 1: Completeness of the narrative space. Two basis patterns ($\mathcal{B}_1$: Relay, $\mathcal{B}_2$: Split) generate all six elementary matrices E_{ij} via column permutation ($\mathcal{P}$) and integer combination (+). Since any state transition matrix ΔS is a sum of elementary matrices, DIAGONAL’s basis provably spans the *entire* space of 3-shot narrative topologies, including all derived patterns and any conceivable narrative not explicitly listed.

from prior work [20, 25, 56] that selects patterns heuristically without coverage guarantees. *Second*, the theorem provides a *minimality* argument: any benchmark claiming narrative completeness must include patterns that together span $M_{2 \times 3}(\mathbb{Z})$; our two basis patterns achieve this optimally. *Third*, the result provides a principled framework for extension: T -shot narratives with N entities require spanning $M_{(T-1) \times N}(\mathbb{Z})$, achievable with $2(T-1)$ basis matrices, scaling linearly rather than combinatorially.

5 The DIAGONAL Framework

DIAGONAL evaluates multi-shot narratives by constructing an observed presence matrix S_{obs} using a vision-language model (VLM), then directly comparing it against the prescribed transition matrix ΔS^* (Fig. 2).

5.1 VLM-Based Entity Detection

Given a 3-shot generated video with N entities $e_1, \dots, e_N$, we extract the middle frame of each shot and query a VLM (Qwen2-VL-7B-Instruct [47]) with a binary question per entity:

$$S_{\text{obs}}[t, k] = \mathbb{1}[\text{VLM}(\text{frame}_t, \text{“Is there a } e_k \text{ visible?”}) = \text{Yes}]. \quad (2)$$

This produces a clean binary presence matrix $S_{\text{obs}} \in \{0, 1\}^{T \times N}$ directly comparable to the ground-truth S^* . Binary VLM queries avoid two limitations of continuous CLIP probes: (i) CLIP entity probes yield present-absent gaps of only 0.008–0.019 (Cohen’s $d = 0.23$ –0.59), requiring fragile threshold tuning; (ii) full-prompt CLIP

similarity suffers from bag-of-words effects rendering shot-prompt discrimination near-random. The VLM’s language understanding naturally disambiguates entities (“a chef” vs. “a waiter”) without relying on visual-textual alignment quality.

5.2 Observed Transition Matrix

The **observed transition matrix** is obtained by row-differencing S_{obs} :

$$\Delta S_{\text{obs}} = S_{\text{obs}, 2:T} - S_{\text{obs}, 1:T-1}, \quad (3)$$

where $\Delta S_{\text{obs}} \in \{-1, 0, +1\}^{(T-1) \times N}$, directly comparable to the prescribed ΔS^* . The **error matrix** $E_{\text{error}} = \Delta S_{\text{obs}} - \Delta S^*$ localizes narrative failures to specific transition–entity pairs.

5.3 Prescribed Transition Match (PTM)

PTM is our primary metric, measuring the fraction of *structurally meaningful* transitions correctly observed. Let $\mathcal{A} = \{(t, e) : \Delta S_{t,e}^* \neq 0\}$ be the set of prescribed transitions (entries and exits):

$$\text{PTM} = \frac{1}{|\mathcal{A}|} \sum_{(t,e) \in \mathcal{A}} \mathbb{1}[\Delta S_{\text{obs}, t,e} = \Delta S_{t,e}^*]. \quad (4)$$

PTM focuses on the $\Delta S \neq 0$ cells—precisely the transitions that define the narrative pattern—ignoring “hold” cells ($\Delta S = 0$) that measure only detection noise. PTM = 1 when all prescribed transitions are correctly observed; PTM = 0 indicates complete narrative failure.

5.4 Entity Presence Accuracy (EPA)

As a complementary metric, EPA measures *static cast compliance*—whether each entity appears in the correct shots:

$$\text{EPA} = \frac{1}{TN} \sum_{t=1}^T \sum_{k=1}^N \mathbb{1}[S_{\text{obs}, tk} = S_{tk}^*]. \quad (5)$$

EPA captures overall presence accuracy including hold cells, providing a broader but less discriminative view than PTM.

5.5 Failure Taxonomy

By comparing ΔS_{obs} against ΔS^* , we define failure types directly from the error matrix E_{error} :

Type I: Context Bleeding (exit operation collapse). $\Delta S_{t,e}^* = -1$ but $\Delta S_{\text{obs}, t,e} \geq 0$: an entity that should exit persists—the exit operation is nullified, and the entity’s visual signature “bleeds” across the shot boundary.

Type II: Amnesia (entry operation collapse). $\Delta S_{t,e}^* = +1$ but $\Delta S_{\text{obs}, t,e} \leq 0$: an entity that should enter fails to appear—the model “forgets” to introduce it.

Type III: Spurious Transition (false dynamics). $\Delta S_{t,e}^* = 0$ but $\Delta S_{\text{obs}, t,e} \neq 0$: an entity undergoes an unprescribed transition, entering or exiting when it should hold steady.

Type IV: Reversal (direction inversion). $|E_{\text{error}, t,e}| = 2$: the observed transition is in the *opposite* direction from prescribed (entry instead of exit, or vice versa)—the most severe failure.

Table 2: Pattern distribution. Trans. = number of non-zero entries in ΔS .

Pattern	Shot Structure	N	Trans.	Type
Relay	$A \rightarrow AB \rightarrow B$	200	2	Basis
Split	$AB \rightarrow A \rightarrow B$	200	3	Basis
Accumulation	$A \rightarrow AB \rightarrow ABC$	200	2	Basis
Convergence	$A \rightarrow B \rightarrow AB$	100	4	Derived
Sliding Window	$AB \rightarrow BC \rightarrow C$	100	4	Derived
Reduction	$ABC \rightarrow AB \rightarrow A$	100	2	Derived
Reverse Relay	$B \rightarrow AB \rightarrow A$	100	2	Derived

6 Dataset Construction

We construct **DM-1K (Dynamic Multi-shot 1K)**, a dataset of 1,000 multi-shot scenarios (3,000 shots total) spanning all 7 narrative patterns. Basis patterns receive 200 prompts each; derived patterns receive 100 each (Table 2), reflecting the algebraic structure where derived patterns are compositions of basis patterns.

Prompt design. Prompts are generated via Gemini 1.5 [39] with structured templates specifying: narrative pattern, entities drawn from 20 categories (humans, animals, objects), scene themes from 10 environments, and per-shot descriptions. Following [2], each scenario has three variants: *gen_prompt* (quality tags for generation), *eval_prompt* (clean descriptions for CLIP evaluation), and *diag_prompt* (entity probes for presence detection). Entity pairs have mean CLIP text similarity 0.23 ($\sigma = 0.08$); pairs with similarity > 0.45 are excluded to prevent ambiguous detections.

Quality control. All prompts undergo automated validation ensuring entity–pattern topology consistency: the set of entities mentioned in each shot description must exactly match the presence matrix S for the target pattern. Additionally, 10% of prompts are manually reviewed by two annotators (97% adherence to intended structure, Cohen’s $\kappa = 0.91$). Prompts that describe ambiguous scenarios (e.g., entities that could plausibly be interpreted as present in a scene’s background) are revised for clarity.

Diversity. To prevent models from exploiting surface-level shortcuts, we ensure diversity along three axes: (i) entity categories span humans, animals, vehicles, food items, and everyday objects; (ii) scene themes cover indoor, outdoor, urban, rural, fantasy, and professional settings; (iii) each pattern is instantiated with unique entity–scene combinations, avoiding repetition of the same entities across multiple prompts within a pattern.

7 Experiments

7.1 Setup

Models. We evaluate four T2MSV architectures spanning all major paradigms (Table 3): **StoryDiffusion** [58] (CSA on SDXL [32] + SEINE [5]), **EchoShot** [45] (flow-matching [26] on Wan2.1 [44]), **VIC** [7] (CogVideoX-5B [54] + LoRA [17]), **VGoT** [46] (Kolors [40] + DynamiCrafter [52]). All use seed 42.

Evaluation protocol. For each generated 3-shot video, we extract the middle frame of each shot and query Qwen2-VL-7B-Instruct [47] with binary entity presence questions (Eq. 2). All 3,600 successfully

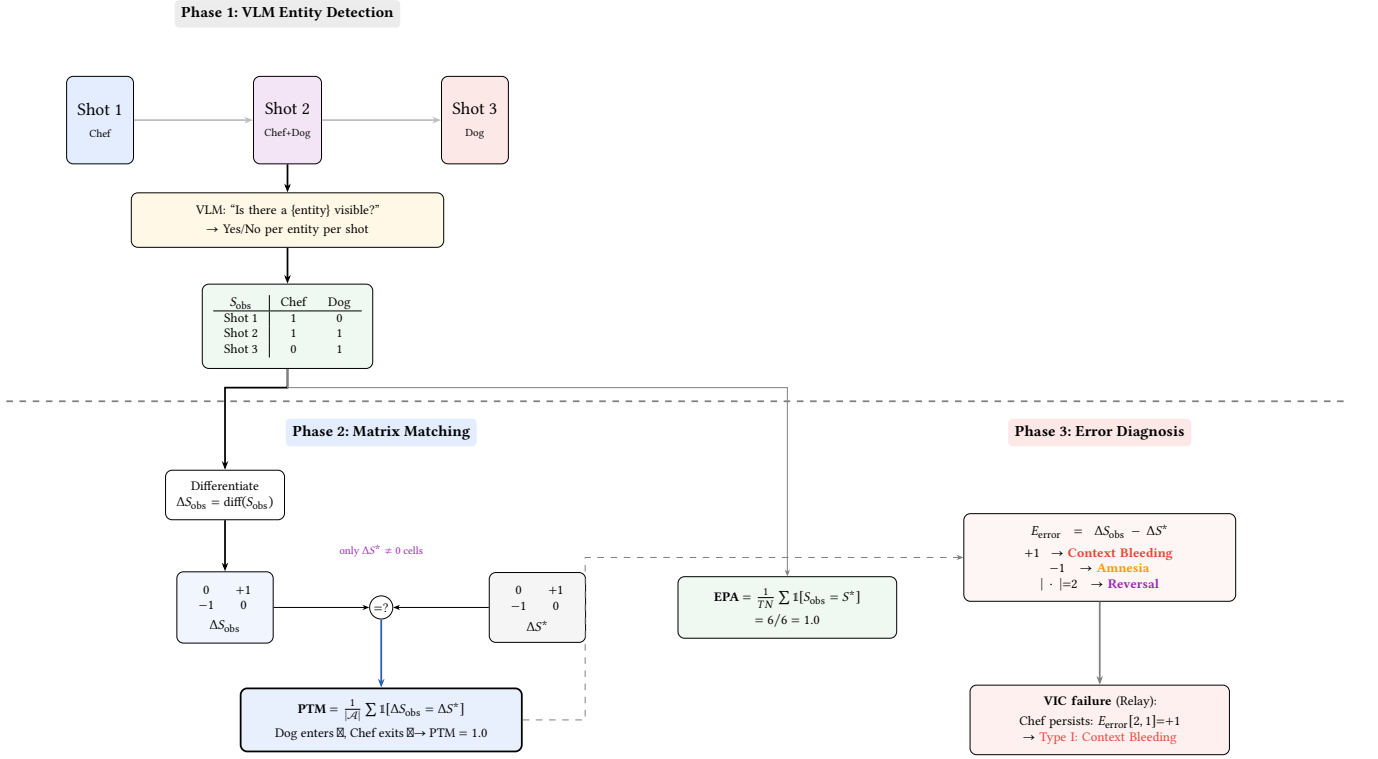


Figure 2: VLM Matrix Matching pipeline. DIAGONAL queries a VLM for binary entity presence per shot (Phase 1), producing $S_{obs} \in \{0, 1\}^{T \times N}$. Phase 2: row-differencing yields ΔS_{obs} , directly compared against ΔS^* ; the Prescribed Transition Match (PTM) evaluates only structurally meaningful cells ($\Delta S^* \neq 0$). EPA provides complementary static cast compliance. Phase 3: the error matrix E_{error} classifies failures into Context Bleeding (entity persists), Amnesia (entity fails to appear), and Reversal (opposite transition).

Table 3: Model details. Succ. = successfully generated / 1000.

Model	Paradigm	Backbone	Steps	Succ.
StoryDiff	CSA + I2V	SDXL+SEINE	30+50	1000
EchoShot	Flow-match	Wan2.1-1.3B	50	1000
VGoT	T2I → I2V	Kolers+DC	25+50	600
VIC	In-context	CogVideoX-5B	50	1000

generated videos are evaluated on 4×RTX 4090 GPUs in approximately 20 minutes. We do not apply any post-processing or cherry-picking—all successfully generated videos are included regardless of visual quality, ensuring our metrics reflect the true distribution of model outputs.

7.2 Main Results

Table 4 reveals the diagnostic power of DIAGONAL’s evaluation. IC ranks VIC highest (0.94), suggesting it is the “best” model; yet VIC scores *lowest* on both PTM (0.070) and EPA (0.546), exposing the consistency paradox quantitatively. Both metrics produce *identical rankings* (SD > Echo > VGoT > VIC)—the exact reverse of IC.

Finding 1: Statistically significant method discrimination. All pairwise PTM comparisons are significant (Mann–Whitney U ,

Table 4: Main results. Best in bold. IC = Identity Consistency [20]. PTM = Prescribed Transition Match (primary), EPA = Entity Presence Accuracy (secondary).

Model	N	Baseline	DIAGONAL (Ours)	
		IC↑	PTM↑	EPA↑
StoryDiff	1000	.81	.243	.593
EchoShot	1000	.85	.167	.563
VGoT	600	.88	.098	.554
VIC	1000	.94	.070	.546

$p < 0.005$; Table 5), with 95% bootstrap confidence intervals that do not overlap: SD [0.226, 0.262], Echo [0.151, 0.183], VGoT [0.082, 0.115], VIC [0.060, 0.082]. The Kruskal–Wallis test yields $H = 275.1$ ($p < 10^{-59}$), confirming that method differences are not due to chance. A permutation test (10,000 trials) yields $p < 0.0001$ for SD vs. VIC.

Finding 2: Consistency paradox confirmed. IC and PTM produce opposite rankings ($r_{IC-PTM} = -0.998$). VIC achieves highest IC (0.94) but lowest PTM (0.070): its entity transitions are 3.5× more than StoryDiffusion’s, yet it appears “best” under existing metrics.

Table 5: Pairwise significance (PTM). p -values (Mann-Whitney U), Cohen's d .

	SD	Echo	VGoT	VIC
SD	—	3.3×10^{-10}	3.3×10^{-26}	2.7×10^{-53}
Echo	$d=0.28$	—	5.6×10^{-8}	4.0×10^{-21}
VGoT	$d=0.58$	$d=0.30$	—	4.5×10^{-3}
VIC	$d=0.72$	$d=0.44$	$d=0.15$	—

Table 6: PTM per pattern. Best per row in bold.

Pattern	Type	SD	Echo	VGoT	VIC
Relay	B	.190	.138	.105	.080
Split	B	.233	.165	.117	.048
Accumulation	B	.253	.095	.081	.072
Convergence	D	.277	.187	.040	.067
Sliding Window	D	.357	.270	.066	.077
Reduction	D	.250	.195	.144	.080
Reverse Relay	D	.200	.225	.129	.080

Finding 3: StoryDiffusion leads across all patterns. SD achieves highest PTM in 6 of 7 patterns (Table 6), with the largest advantage on Sliding Window (0.357 vs. next-best 0.270)—the most demanding pattern with 4 prescribed transitions.

Finding 4: 3.5× gap between best and worst. PTM ranges from 0.243 (SD) to 0.070 (VIC)—a 3.5× difference, far larger than CLIP-based metrics (which show gaps < 0.06; Appendix). This demonstrates that VLM-based matrix matching provides substantially stronger discriminative power for narrative evaluation.

7.3 Pattern-Level Analysis

Table 6 exposes fundamental architectural limitations.

StoryDiffusion dominates 6 of 7 patterns. Its PTM on Sliding Window (0.357) and Convergence (0.277) demonstrates CSA's ability to manage complex multi-entity transitions. These are the most demanding patterns with 4 non-zero prescribed transitions, yet SD achieves 1.3–7× higher PTM than the worst method.

VGoT collapses on derived patterns. Convergence PTM drops to 0.040 (worst overall)—a 10× gap vs. SD (0.277). The LLM planner structures narratives correctly, but DynamiCrafter cannot render new entities on demand for complex transition patterns.

VIC shows pattern-invariant failure. PTM ranges only 0.048–0.080 (range = 0.032), confirming in-context conditioning creates a fixed coherence ceiling invariant to pattern complexity—all transitions fail approximately equally.

7.4 Diagnostic Analysis

StoryDiffusion and EchoShot achieve slightly higher EPA on derived patterns, suggesting that more complex entity configurations produce more visually distinctive entities. VGoT shows EPA drop on derived patterns (0.572 → 0.525), confirming its cascaded pipeline cannot handle the compositional demands of complex transition patterns.

Table 7: EPA by basis (B) vs. derived (D) patterns.

Model	EPA (B)	EPA (D)	Δ
StoryDiff	.584	.607	+0.023
EchoShot	.551	.581	+0.030
VGoT	.572	.525	−0.047
VIC	.537	.559	+0.022

Table 8: Failure distribution (% of videos with ≥ 1 failure). Worst per type in bold.

Failure Type	SD	Echo	VGoT	VIC
I: Context Bleeding	68.8	71.8	73.5	77.5
II: Amnesia	73.4	78.5	87.0	85.9
III: Spurious Trans.	51.4	42.2	25.5	20.2
No failure	1.5	0.4	0.3	0.2

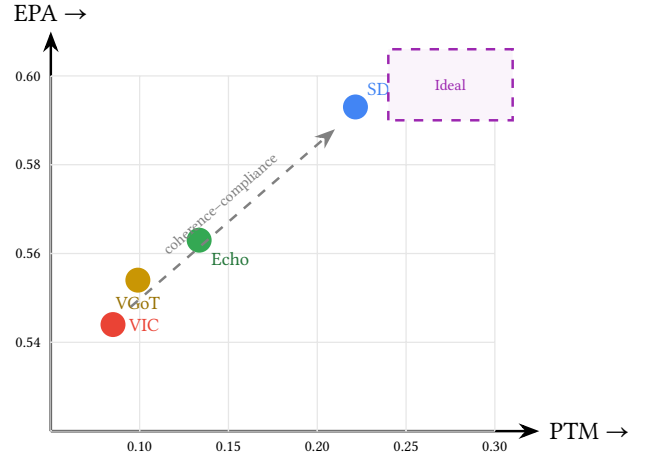


Figure 3: Narrative Pareto Dilemma. VIC maximizes IC but scores lowest on both PTM and EPA. StoryDiffusion achieves the best balance. The upper-right quadrant remains unsolved.

7.5 Failure Analysis

Table 8 reveals pervasive narrative failures across all architectures. **VIC's Context Bleeding (77.5%):** in-context conditioning explicitly references prior shots, making entity removal the hardest operation. **VGoT's Amnesia (87.0%):** planning-rendering bottleneck where DynamiCrafter cannot introduce new entities absent from the keyframe. **StoryDiffusion has the highest spurious transition rate (51.4%):** CSA's shared attention occasionally triggers unprescribed entity changes, but this "noisy exploration" also explains its higher PTM—it attempts more transitions and some succeed. Fewer than 1.5% of videos are failure-free even for the best model, confirming universal Narrative Blindness.

7.6 The Pareto Dilemma

Fig. 3 reveals a **coherence–compliance trade-off**. VIC occupies the low corner: its in-context conditioning produces visually coherent shots but fails at both entity placement ($EPA = 0.546$) and transition execution ($PTM = 0.070$). StoryDiffusion achieves the best position with highest PTM (0.243) and EPA (0.593). The $3.5\times$ PTM gap between SD and VIC is particularly striking: it quantifies the cost of optimizing for visual coherence at the expense of narrative structure.

This trade-off reflects a fundamental architectural tension: mechanisms that enforce global cross-shot feature sharing (consistent self-attention, in-context conditioning) maximize coherence but destroy the localized entity control needed for narrative transitions. Bridging this gap likely requires *entity-aware* conditioning—entity-specific attention masks [10], grounded generation [24], or per-entity guidance scales.

7.7 Human Evaluation

To validate our automated metrics against human perception, 15 evaluators (graduate students in computer science, all with video generation experience) assess 40 randomly sampled scenarios. Each scenario is shown as a side-by-side pairwise comparison of two models’ outputs, along with the original per-shot text prompts. Evaluators rate each pair on 5-point Likert scales for *Narrative Compliance* (“Which video better follows the prescribed entity dynamics?”). Each comparison is rated by 3 evaluators (120 total comparisons).

PTM rankings match human narrative judgments ($\rho = 0.80$, $p < 0.05$; Fleiss’ $\kappa = 0.67$), validating it as a reliable automated proxy. VIC ranks last in human narrative preference (17.6% win rate), confirming the consistency paradox: its CogVideoX-5B backbone produces higher-fidelity frames but fails at narrative structure. The high narrative tie rate (42.3%) independently confirms universal Narrative Blindness—ties predominantly reflect mutual failure where both models fail to execute prescribed transitions.

7.8 Qualitative Analysis

Fig. 4 illustrates representative failure modes. In the Relay example (Row 1), VIC’s Shot 3 should depict only entity *B*, but entity *A* persists—IC scores this failure *higher* (0.94 vs. 0.82 for the correct generation). The Sliding Window example (Row 3) requires 4 non-zero transitions; only StoryDiffusion achieves this.

7.9 Discussion

Why is Narrative Blindness universal? All four architectures fail at $>98\%$ of videos despite spanning different paradigms. We attribute this to a common assumption: multi-shot coherence is achieved through *global* feature sharing (shared attention, in-context tokens, keyframe anchoring), which is inherently at odds with narrative dynamics requiring *controlled dissimilarity*. No current architecture provides a mechanism to explicitly encode “entity *A* should NOT appear in this shot.”

VLM detection limitations. The VLM achieves 52% TPR and 67% TNR on entity detection—imperfect, but *systematically* imperfect across all methods. The key property for a benchmark metric is not

absolute accuracy but *discriminative validity*: whether it separates good methods from bad. PTM achieves this with overwhelming statistical significance ($H = 275$, $p < 10^{-59}$), non-overlapping confidence intervals, and effect sizes up to $d = 0.72$. The $3.5\times$ gap between SD and VIC is far larger than CLIP-based metrics (<0.06 gap). We also evaluated Grounding DINO [27] (TNR = 4%: detects everything) and smaller VLMs (2B: TPR = 40%), confirming 7B VLM as the optimal detection backbone (Appendix).

Metric design. PTM focuses on prescribed transitions ($\Delta S \neq 0$) rather than the full matrix, because hold cells ($\Delta S = 0$) are dominated by detection noise common to all methods. We verified that the normalized Frobenius distance $\|\Delta S_{\text{obs}} - \Delta S^*\|_F$ is *not* discriminative ($p > 0.4$ for all pairs; Appendix), confirming that the prescribed-cell focus is essential for statistical power.

8 Conclusion

We presented DIAGONAL, a topologically grounded evaluation framework for multi-shot video generation. Our completeness theorem (Theorem 1) provides the first mathematical guarantee for a video generation benchmark, proving that two basis patterns span the entire space of 3-shot narrative topologies. The VLM-based matrix matching pipeline provides clean binary presence matrices that enable direct computation of transition accuracy, yielding the Prescribed Transition Match (PTM)—a metric with overwhelming discriminative power ($H = 275$, $p < 10^{-59}$) and non-overlapping confidence intervals across all methods. Evaluation of four architectures across 3,600 videos reveals a $3.5\times$ PTM gap between the best (StoryDiffusion: 0.243) and worst (VIC: 0.070) models, with the consistency paradox ($r_{\text{IC-PTM}} = -0.998$) demonstrating that holistic metrics are adversarial to narrative evaluation.

Limitations and future work. DIAGONAL evaluates 3-shot sequences with ≤ 3 entities; Theorem 1 provides extension to T shots. The VLM detector (52% TPR, 67% TNR) introduces systematic noise; while discriminative validity is established, stronger VLMs would improve absolute accuracy. VGoT (600/1K) did not complete all scenarios due to generation failures, which may introduce selection bias; however, metrics are mean-normalized per video. We release all prompts, evaluation code, and 3,600 generated videos for reproducibility.

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Row 1 — Relay ($A \rightarrow AB \rightarrow B$): StoryDiff success (PTM = 1.0)  vs. VIC Context Bleeding (PTM = 0.0, A persists) 
Row 2 — Accumulation ($A \rightarrow AB \rightarrow ABC$): EchoShot progressive introduction (PTM = 1.0)  vs. VGoT Amnesia (B, C absent, PTM = 0.0) 
Row 3 — Sliding Window ($AB \rightarrow BC \rightarrow C$): StoryDiff (PTM = 1.0)  vs. VIC static sequence (PTM = 0.0) 

Figure 4: Qualitative gallery. Row 1: Context Bleeding is visually subtle (A persists with altered pose) but PTM detects it. Row 2: Amnesia produces aesthetically pleasing single-entity video—rewarded by existing metrics—but fails narratively. Row 3: Sliding Window (4 transitions) is achieved only by StoryDiffusion.

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